

This question paper contains 4 printed pages.

Your Roll No.

No. of Paper : 6632 HC
Unique paper code : 32351401
Name of the paper : Partial Differential Equations
Name of course : B.Sc. (Hons.) Mathematics
Semester : IV
Duration : 3 hours
Maximum marks : 75

*(Write your Roll No. on the top immediately
on receipt of this question paper.)*

*All Sections are compulsory.
Marks of each part are indicated.*

SECTION – I

Attempt any two parts out of the following.

(a) Obtain the general solution of the equation

$$(z^2 - 2yz - y^2)p + x(y + z)q = x(y - z).$$

Find the integral surfaces of this equation passing through (i)
the x-axis, (ii) the y-axis, (iii) the z-axis.

$$\left(7\frac{1}{2}\right)$$

(b) Solve the following initial value system:

$$u_t + uu_x = e^{-x}v, \quad v_t - av_x = 0$$

$$\text{with } u(x,0) = x \quad \text{and} \quad v(x,0) = e^x.$$

$$\left(7\frac{1}{2}\right)$$

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- (c) Apply the method of separation of variables $u(x, y) = f(x)g(y)$ to solve

$$x^2 u_{xy} + 9y^2 u = 0 \quad u(x, 0) = \exp\left(\frac{1}{x}\right) \quad \left(7\frac{1}{2}\right)$$

SECTION – II

Attempt any one part out of the following.

2. (a) Show that the equation of motion of a one-dimensional wave equation is $u_{tt} = c^2 u_{xx}$ where $c^2 = \tau/\rho$.
- (b) Derive the Laplace equation of motion.

Attempt any two parts out of the following.

3. (a) Find the characteristics, characteristic coordinates and reduce the equation given below to the canonical form:

$$u_{xx} + (\sec h^4 x) u_{yy} = 0.$$

- (b) Transform the following equations to the form:

$$v_{\xi\eta} = cv, \quad c = \text{constant}, \quad u_{xx} - u_{yy} + 3u_x - 2u_y + u = 0$$

by introducing the new variable $v = ue^{-(a\xi + b\eta)}$, where a and b are undetermined coefficients.

- (c) Determine the general solution of the equation given below:

$$3u_{xx} + 10u_{xy} + 3u_{yy} = 0$$

SECTION – III

Attempt any three parts out of the following.

4. (a) Determine the solution of the equation:

$$L[u] \equiv u_{xy} + au_x + bu_y + cu = f(x, y)$$

where a, b, c and f are differentiable functions of (x, y) in some domain D^* , under the boundary conditions that u and u_x are prescribed along the curve C in the xy plane. (7)

(b) Solve the Goursat problem

$$xy^3 u_{xx} - x^3 y u_{yy} - y^3 u_x + x^3 u_y = 0, \quad x \neq 0,$$

$$u(x, y) = f(x) \quad \text{on} \quad y^2 + x^2 = 16 \quad \text{for} \quad 0 \leq x \leq 4,$$

$$u(x, y) = g(y) \quad \text{on} \quad x = 0 \quad \text{for} \quad 0 \leq y \leq 4,$$

$$\text{where} \quad f(0) = g(4). \quad (7)$$

(c) Find the solution of the initial boundary value problem:

$$u_{tt} - c^2 u_{xx} = 0$$

$$u(x, 0) = \sin x, \quad u_t(x, 0) = x^2 \quad (7)$$

(d) Solve the Cauchy problem for the non-homogeneous wave equation

$$u_{tt} = c^2 u_{xx} + e^x,$$

with the initial conditions:

$$u(x, 0) = 5, \quad u_t(x, 0) = x^2 \quad (7)$$

SECTION – IV

Attempt any **three** parts out of the following.

(a) Solve using the method of separation of variables:

$$u_t = k u_{xx}, \quad 0 < x < l, \quad t > 0.$$

$$u(0, t) = 0, \quad t \geq 0$$

$$u(l, t) = 1, \quad t \geq 0$$

$$u(x, 0) = \sin \frac{\pi x}{2l}, \quad 0 \leq x \leq l. \quad (7)$$

(b) Using the method of separation of variables, discuss problem of vibrating string:

$$u_{tt} - c^2 u_{xx} = 0, \quad 0 < x < l, \quad t > 0,$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x), \quad 0 \leq x \leq l,$$

$$u(0, t) = u(l, t) = 0, \quad t > 0.$$

(c) Determine the solution of Initial Boundary Value problem

$$u_{tt} = c^2 u_{xx} + x^2, \quad 0 < x < 1, \quad t > 0,$$

$$u(x, 0) = x, \quad u_t(x, 0) = 0, \quad 0 \leq x \leq 1,$$

$$u(0, t) = 0, \quad u(1, t) = 1, \quad t > 0.$$

(d) Find the solution of the plucked string of length l equation given by:

$$u_{tt} - c^2 u_{xx} = 0$$

$$u(x, 0) = 0,$$

$$u_t(x, 0) = \begin{cases} \frac{v_0 x}{a}, & 0 \leq x \leq a \\ \frac{v_0 (l-x)}{(l-a)}, & a \leq x \leq l \end{cases}$$

$$u(0, t) = u(l, t) = 0, \quad t > 0.$$